HPCME Systems LLC
Single Business Tower, Office 05, 31st Floor
Business Bay, Sheikh Zayed Road
PO Box – 36677, Dubai UAE
Tel : 04 507 2015
Email : {sales,info}@hpcme.com
VAT No. : 100440073300003 Company Reg. No. : 910876

Support Service Level Agreement (SSLA)
Khalifa University, Abu Dhabi

June 26, 2022

Location of Work: Khalifa University, Abu Dhabi

Support Hours (UAE Time Zone):

Weekdays: 9:00- 17:00 with response time 4 hours

After hours: 17:00-8:00 (following day): response time 8 hours

Weekend (Friday 17:00)-Monday 9:00 AM: response time 8 hours

Contractor: HPCME SYSTEMS LLC

Subject: HPC Cluster Software Maintenance

Support Period: One calendar year from date of PO.

Scope Of Support

1 Communication

All support requests must be sent to the official support emails:

support@hpcme.com or/and support@q-leap.com

2 Support and Maintenance

This contract will be for one calendar year from date of the purchase order.

This contract shall cover:

a. Support for integration between Qlustar storage system
   i. Includes storage and Qlustar LDAP integration
b. Monitoring with Nagios/ Ganglia. This includes, but not limited to:
   i. Addition of performance monitoring for Lustre file system monitoring
   ii. Compute node IB performance.
   iii. Monitoring Network & Mellanox Switches
c. Cluster maintenance
   i. Cluster malfunction
   ii. Cluster performance fine-tuning
   iii. Cluster repair in case of replacement of faulty hardware
   iv. Bug fixes of software (Lustre, QLuman, SLURM, OS)
d. Cluster security patches
   i. This includes urgent security patches
e. Root cause analysis and reporting of issues
f. Minor OS version updates
g. Software support for SLURM, Lustre, Qluman, OS
h. Software upgrades such as SLURM, Lustre, Qluman, OS

3 Technical Cluster Change

a. Any change must be ratified by the stakeholders (Khalifa University, HPCME, QLeap).
b. Any change that compromises the stability of the cluster is the sole responsibility of the subject that made the change.
c. Any POC is billable.
d. Any change that is deemed minor and requires less than 45 minutes shall be treated as support issue. Examples are, installing new software, changing DNS entries, Adding/removing users and some minor changes in SLURM .
e. Any change that requires more time requires a process to assess the risks of the change on the stability of the cluster. This will be considered as billable time.
f. A balance of 100 hours of development from the original software developer are included this contract

g. At the successful implementation of the a given change a formal email will be sent to keep track of the number of hours spent and remaining balance and an email request will be sent requesting purchase of additional time.

4 Out of Scope

a. This contract does not cover hardware support such as replacement of faulty hardware.
b. Upon identifying the hardware failure, any software malfunction is not covered if the failed hardware is not addressed in a timely fashion.
c. Failures due to environmental issues (for example cooling/power issues) compromising the stability of the system.

5 Roles & Responsibilities

5.1 HPCME Roles & Responsibilities

a. Providing high quality services as specified in the scope and ensuring periodic review of the service performance.
b. Meeting the performance levels and targets mentioned in this SLA.
c. Providing regular reports and updates to KU regarding the SLA adherence.

5.2 KU Roles & Responsibilities

a. Providing access on campuses and resources to facilitate HPCME and its subcontractor's staff performing the maintenance and/or support.
b. Reporting any issue or service request using the agreed communication channels

6 Support Levels and Targets

6.1 Prioritization in Incident Handling

Priority Definitions:

A set of criteria will be used to prioritize incidents considering both the urgency of the incident (how quickly the customer needs the resolution) and impact (the level of business impact it is causing).

The following matrix shows the priority levels of the incidents based on the impact and urgency:

		Urgency		
		High	Medium	Low
Impact	High	Priority 1 (Critical)	Priority 2 (High)	Priority 3 (Medium)
	Medium	Priority 2 (High)	Priority 3 (Medium)	Priority 4 (Low)
	Low	Priority 3 (Medium)	Priority 4 (Low)	Priority 4 (Low)

Definition of Urgency:

Urgency is an indication of how quickly the customer needs the resolution of the incident. Urgency is assessed as follows:

Urgency Value	Definition
High	End users are unable to complete their work, and There is no viable workaround, and Service restoration must be completed immediately or else significant negative impact of productivity will occur.
Medium	End users are unable to complete their work, and End users cannot operate at the most optimum level.
Low	End users are able to complete their work using a workaround, or End users are unable to complete their work using a workaround. However, service restoration can be delayed without significant impact of productivity to meet operational goals.

Definition of Impact:

Impact is the effect on the business due to the loss of the service.
Impact is assessed as follows:

Impact Value	Definition
High	A service is completely down, or Service interruption affects multiple users in different departments.
Medium	A service is partially impacted, or Localized service interruption affecting users in a single department.
Low	Service interruption affects a few users.

6.2 Targets for Incidents Handling

Priority	Response Time Working Hours	Resolution Time Working Hours (*)
Critical	30 minutes	4 hours
High	1 hour	8 hours
Medium	4 hours	16 hours
Low	6 hours	24 hours

(*) Exceptions: HPCME will not be liable for any failure or disruption of services in the following cases:

- "**Force Majeure Event**" that includes any cause beyond the reasonable control of HPCME that could not be avoided, such as acts of war, earthquakes and other circumstances.
- Planned outage.
- Outage is due to hardware failure.
- Upon identifying the hardware failure, any software malfunction is not covered if the failed hardware is not addressed in a timely fashion.
- Failures due to environmental issues (for example cooling/power issues) compromising the stability of the system.
- Working Hours are indicated in section at start of document.

6.3 Service Maintenance

- HPCME will perform regular service maintenance which might cause service interruptions.
- **Planned Outage:** For maintenance activities that require downtime/service interruption, HPCME will communicate to KU at least five business days before the planned implementation time. The notification will be sent via email, and will include a detailed description of the activity and the expected duration of the downtime/service interruption. An exception can be made for critical security updates and patches.
- **Unplanned Outage:** in case of any unplanned outage/service interruption, HPCME will send a notification to KU via email, which include the details of the outage and expected resolution time. Another notification will be sent once the services are restored.

6.4 Service Reporting & Communications

No.	Type (Report/Meeting)	Details	Frequency
1	Report	Tickets Workload, Resolution Progress and Service Level Adherence	Quaterly
2	Meetings	Quarterly or as required by operational needs	Quarterly or as required

6.5 Failure to adhere to SLA

If HPCME fails to perform or is late in responding then the following procedure will be followed:

1. If HPCME did not respond within the targets outlined above on items included in the scope of the service, then KU will provide a first reminder to HPCME.
2. If HPCME fails to respond to the reminder within 8 working hours. This will be classed as failure to adhere to the SLA, and KU will send a second reminder. This will incur penalty of 1% of the total SLA value.
3. If the HPCME fails to respond after working 3 days from the first reminder, this will be classed as a further failure to adhere to the SLA. This will incur penalty of 3% of the total SLA value.
4. The penalties will capped to maximum of 20% of the total value of the SLA.

7 Commercial Offer

ID	Description	Qty	Unit Price (AED)	VAT 5%	Total (AED)
SUP	Extended Support for Al Mesbar HPC Cluster	12 months	36,750.00	1,837.5	463,050.00
Customization	100 hours of extra customizations that are outside of support of as per built cluster	1	110,000	5,500.00	115,500.00
Total					578,550.00